Statement of Purpose

Title: OWASP AI Model Watermarking.

Problem Statement: To develop a secure and robust AI model watermarking system that enables verifiable ownership authenticationwhileprotecting AI models against theft, tampering, and unauthorized use.

Objectives:

Semester 7:

- Develop a secure cryptographic watermark embedding mechanism for AI/ML models across different frameworks (TensorFlow, PyTorch, ONNX).
- Implement a privacy-preserving Zero-Knowledge Proof (ZKP) generation and verification scheme (using zk-SNARKs via Circom and SnarkJS) for model ownership authentication.
- Design and build a core open-source, framework-agnostic Python SDK/API for seamless watermark integration and detection.

Semester 8:

- Evaluate and enhance watermark robustness against adversarial model manipulation attacks, including fine-tuning, weight pruning, quantization, model compression, and model extraction.
- Develop a web-based dashboard (React.js, FastAPI, Tailwind CSS) for model provenance tracking, deployment monitoring, and audit-ready compliance reporting.
- Perform end-to-end integration, performance evaluation (balancing watermark resilience vs. model accuracy), and comprehensive documentation ofthe system.

Domain: Cybersecurity · Artificial Intelligence / LLMs.

Methodology Used:

Theproposed methodology employs a multi-layered security framework for AI model IP protection. First, a cryptographic watermark is embedded into the neural network's weights, biases, or activation boundaries during or post-training without degrading model performance. Second, to verify ownership without exposing the watermark key, a Zero-Knowledge Proof (ZKP) pipeline is established: the model owner generates a zk-SNARK proof using a cryptographic relation, which a third-party verifier can validate off-chain or on-chain using public keys, ensuring zero leakage of the watermark. Third, the system's robustness is systematically tested against pruning, quantization, and fine-tuning to ensure resistance. Finally, a web-based management dashboard interacts with a FastAPI backend and a PostgreSQL database to manage watermark tracking, deployment logs, and verification history, enabling secure model provenance and regulatorycompliance.

Tools Used:

- **Programming Languages:** Python, JavaScript, HTML5, CSS3, Circom (for ZK circuits)
- **AI / Machine Learning:** TensorFlow, PyTorch, NumPy, Scikit-learn, ONNX
- **Cryptography & Security:** SHA-256, RSA / ECC, Circom, SnarkJS (zk-SNARK)
- **Web Development & APIs:** React.js, Tailwind CSS, FastAPI / Flask
- **Database&Storage:** PostgreSQL / MongoDB

Expected Outcomes:

- A structured, framework-agnostic Python SDK and API for embedding and detecting watermarks in various formats (.pt, .h5, .onnx).
- A working, privacy-preserving ZKP-based verification pipeline that validates ownership without revealing the embedded watermark weights or signatures.
- An interactive, responsive web dashboard for visual model management, verification logging, and compliance auditing.
- An adaptive security layer that maintains high watermark extraction rates even after severe model pruning, quantization, or fine-tuning attacks.

Role Distribution:

- **Sami Daire(231813) - Project Lead & Cryptography & Verification Developer:** Responsible for designing the cryptographic ZKP circuits in Circom, setting up SnarkJS verification, and managing overall project coordination and system integration.
- **Mohamed Ansari (231803) - AI & Watermarking Developer:** Responsible for researching and implementing watermark embedding algorithms in PyTorch/TensorFlow, developing detection mechanisms, and conducting robustness testing against pruning, quantization, and fine-tuning.
- **Arfat Shaikh (231846) - Frontend & Integration Developer:** Responsible for implementing the React.js dashboard, integrating it with the backend APIs, and packaging the Python SDK/API.
- **Yash Bandiwadekar (231810) - Backend Developer:** Responsible for developing the FastAPI backend, designing the PostgreSQL database, and building the REST APIs for watermark tracking and verification history.

Department Review Committee

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